Figure 12.4 shows the basic URL where http is a protocol, www.cms.tn.gov.in is a hostname / domain name, sites/default/files/press_release are the folders and pr070119a.jpg is the file name. These are all the basic information which can be obtained from a URL.

URL Type

Depending on the location of the document the URL is divided into 2 types

- Absolute URL
- Relative URL

12.4.1 Absolute URL

Absolute URL is the complete address of a document on the Internet. Absolute URL contains all the information that are required to find the files on the Internet. These are similar to postal address if any of the information is missing then the post will not be delivered to the right person. Similarly, if any of the four parts is missing then the browser would not able to link to the specific file. So, all the four parts is very important in absolute URL.

12.4.2 Relative URL

Relative URL is the partial address of a document on the Internet. Relative URL contains only file name or file name with folder name. We can use this type of URL when the file is on the same server related to original document.

12.5 DNS Components

There are three important components in the Domain Name System. They are

- Namespace
- Name server
- Zone

12.5.1 Name Space

The domain names must be very unique and appropriate. The names should be selected from a names pace. The name space can be organized in two ways

- Flat name space
- Hierarchical name space

Flat name space is where the name is assigned to the IP address. They do not have any specific structure. In this flat name space, some meaningful names are given to IP address for accessing. The major disadvantage of flat name space is that they cannot be used in large system. Because they need to be accessed and controlled centrally to avoid ambiguity and redundancy. But it is difficult in flat name system. To avoid this major disadvantage hierarchical name space is used in large.

Hierarchical name space is where the name is made up of several parts. The first part may represent the nature of organization, the second part may represent the name of organization, and third part may represent the department of the organization and so on. In this way the power to control the name space can be decentralized.

Domain Name Space

Domain name space was designed to achieve hierarchical name space. In this, the names are represented as a tree like structure with root element on the top and this tree can have a maximum of 128 levels starting from root element taking the level 0 to level 127.

XII CompApp_English version CHAPTER 12.indd 165 23-02-2020 12:45:43